



 MAJOR PROJECT FOR 21 DAYS SQL CHALLENGE

SQL Murder Mystery

“Who Killed the CEO?”

Indian Data Club

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 CASE FILE #2025-OCT

The Incident

Date: October 15, 2025 | 21:00 Hours

Victim: TechNova Inc. CEO

Location: CEO Office, Top Floor

The CEO was found dead. As the Lead Data Analyst, you must navigate the company's secure SQL databases to solve the case. The answers lie in the data.



STEP 01: EVIDENCE_LOGS

Where & When?

We begin by querying the evidence table to lock down the precise time and location of the body's discovery.

```
-- Identify where and when the crime happened
SELECT room, found_time
FROM evidence
WHERE room LIKE 'CEO Office'
ORDER BY found_time
LIMIT 1;
```



Who Entered?

Analyzing `keycard_logs` reveals every employee who physically entered the CEO's office during the murder window (20:30 - 21:00).

```
-- Analyze who accessed critical areas
SELECT e.name, e.role, kl.room,
       kl.entry_time, kl.exit_time
FROM keycard_logs kl
INNER JOIN employees e
  ON kl.employee_id = e.employee_id
WHERE kl.room = 'CEO Office'
  AND kl.entry_time BETWEEN
    '2025-10-15 20:30:00'
  AND '2025-10-15 21:00:00'
ORDER BY kl.entry_time;
```

Checking Alibis

We cross-reference their stated alibis against the hard data in the logs. If they claimed to be elsewhere but badge data says otherwise, we have a suspect.

```
-- Cross-check alibis with actual logs
SELECT e.name, kl.employee_id,
       kl.room AS actual_location,
       a.claimed_location AS story_location,
       a.claim_time
FROM alibis a
JOIN keycard_logs kl
     ON a.employee_id = kl.employee_id
INNER JOIN employees e
     ON e.employee_id = a.employee_id
WHERE a.claim_time BETWEEN
       kl.entry_time AND kl.exit_time
AND a.claimed_location ≠ kl.room;
```



Suspicious Calls

Isolating communication records. We filter the `calls` table for any activity happening exactly around the time of death.

```
-- Investigate suspicious calls
SELECT e.name AS caller,
       e2.name AS receiver,
       c.caller_id, c.receiver_id,
       c.call_time, c.duration_sec
FROM calls c
JOIN employees e
  ON c.caller_id = e.employee_id
JOIN employees e2
  ON c.receiver_id = e2.employee_id
WHERE c.call_time BETWEEN
      '2025-10-15 20:40:00'
      AND '2025-10-15 21:00:00';
```

The Lie Detector

Using a SQL CASE statement to programmatically label employee statements as 'Truth' or 'LIE DETECTED' based on the discrepancy found.

```
-- Match evidence with movements and claims
SELECT e.name, a.claimed_location AS claim,
       kl.room AS actual,
       CASE
         WHEN a.claimed_location = kl.room
         THEN 'Truth'
         ELSE 'LIE DETECTED'
       END AS status
FROM employees e
JOIN alibis a ON ...
JOIN keycard_logs kl ON ...
ORDER BY status ASC;
```

```
12:01a 23- 1
C2 30 REP #$30
18 CLC
F8 SED
49 34 12 LDA #$1234
59 21 43 ADC #$4321
BF 03 7F 01 STA $017F03
08 CLD
E2 30 SEP #$30
00 BRK
```

```
NUMxDIZC .A .X .Y SP DP DB
00110000 0000 0000 0002 CFFF 0000 00
```

```
NUMxDIZC .A .X .Y SP DP DB
00110000 5555 0000 0002 CFFF 0000 00
```

3

```
55 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00:00
```



FINAL_VERDICT

Case Closed

The intersection of all datasets points to one individual. They were in the room, made a call during the crime, and falsified their alibi.

```
-- Combine all findings to identify the killer
SELECT e.name AS killer
FROM employees e
JOIN keycard_logs kl ON ...
JOIN calls c ON ...
JOIN alibis a ON ...
WHERE kl.room = 'CEO Office'
      AND c.call_time BETWEEN ...
      AND a.claimed_location ≠ kl.room;
```

Mission Accomplished



DATA EXTRACTION

Retrieved critical evidence
from relational databases.



COMPLEX LOGIC

Mastered JOINS and Logic
Gates to filter suspects.



JUSTICE SERVED

The killer has been
identified and processed.



Image Sources



<https://ahouseinthehills.com/wp-content/uploads/2025/01/Dark-Office-Ideas-for-a-Stylish-Moody-Workspace-.jpg>

Source: ahouseinthehills.com



https://www.starwitness.com/uploads/2/1/6/0/21605310/monitor-with-processing-screen_orig.png

Source: www.starwitness.com



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Image Sources



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